

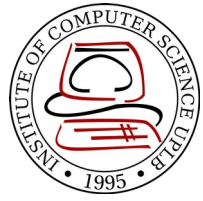
CMSC 137

Data Communications and Networking

Lecture Module
07 – Digital Transmission



University of the Philippines
LOS BAÑOS



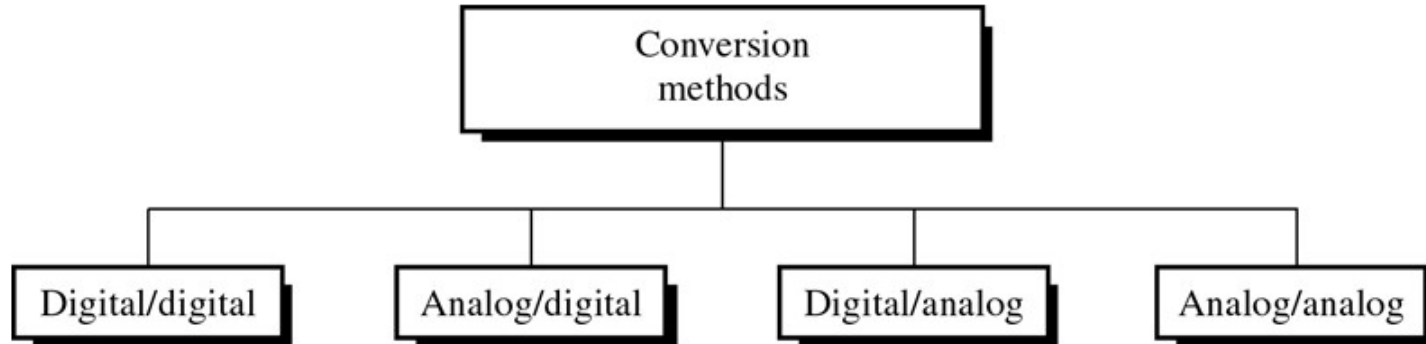
Module 7: Digital Transmission

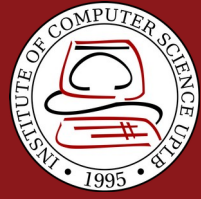
Rozano S. Maniaol
Lecturer

Introduction



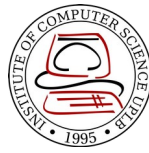
A computer network is designed to send information from one point to another. The information needs to be converted to either digital signal or analog signal for transmission.



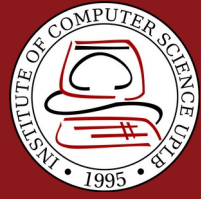


Digital to Digital Conversion

Digital to Digital Conversion

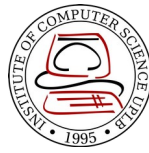


- Represent digital data by using digital signals
- Involves 3 techniques:
 - Line Coding
 - Block Coding
 - Scrambling
- Line coding is always needed; block coding and scrambling may or may not be needed



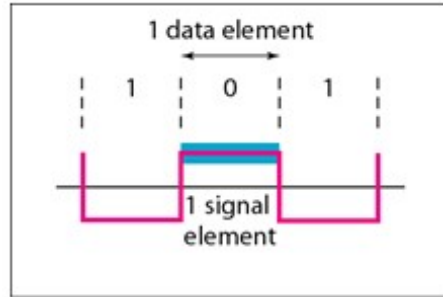
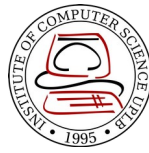
Line Coding

Definition

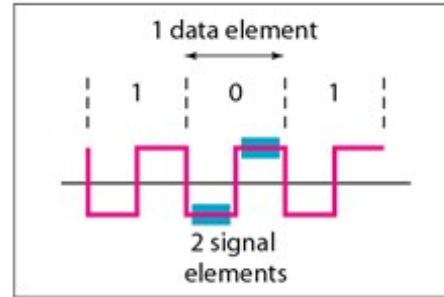


- Data Element
 - Smallest entity that can represent a piece of information (bit)
- Signal element
 - Shortest unit (timewise) of a digital signal
- Ratio “r” is the number of data elements carried by each signal element

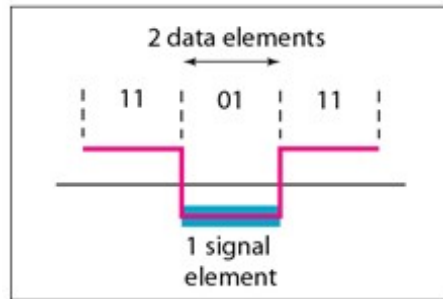
Signal Element vs Data Element



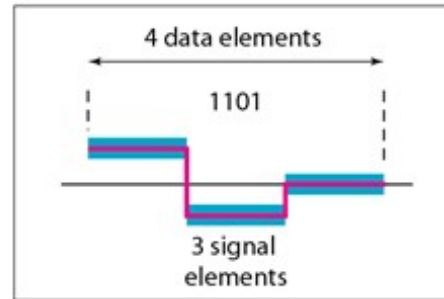
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)

Definition



- **Data Rate** (Bit Rate)
 - Number of bits sent in 1 second (bps)
- **Signal Rate** (Pulse Rate, Modulation Rate, Baud Rate)
 - Number of signal elements sent in 1 second (baud)
- Goal: Increase data rate (**increase speed**) while decreasing signal rate (**decrease bandwidth requirement**)

$$S = c \times N \times \left(\frac{1}{r}\right)$$

N-data rate

c-case factor

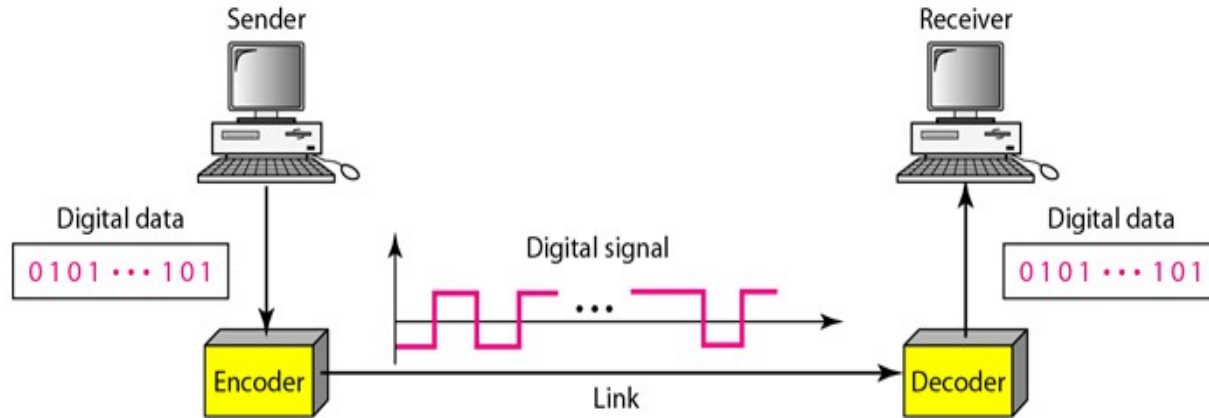
S-number of signal elements

r-ratio of data element per signal element

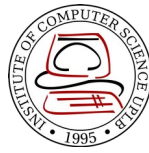
Line Coding



Process of converting digital data (strings of 0s and 1s) into digital signal

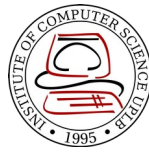


Considerations for choosing good coding scheme



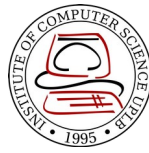
- Effective Bandwidth
 - Finite compared to Absolute Bandwidth
 - Baud rate determines the required bandwidth
 - For now, Bandwidth is proportional to Baud Rate

Considerations for choosing good coding scheme



- Baseline Wandering
 - Receiver calculates a running average of the received signal power – baseline
 - Incoming signal is evaluated against baseline (to determine 1 or 0)
 - Long sequence of 0's or 1's can cause a drift in the baseline
 - Line coding scheme must prevent this

Considerations for choosing good coding scheme

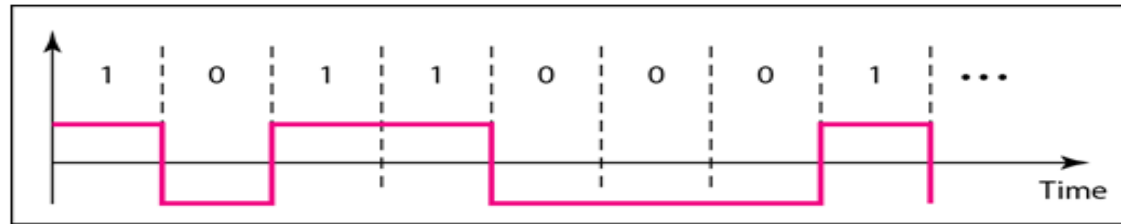


- **Direct Current** (DC) Components
 - Very low (near zero) frequencies created by the spectrum when digital signal is constant for a while
 - Some systems cannot pass low frequencies
 - Some systems use electrical coupling
 - No DC component is desired

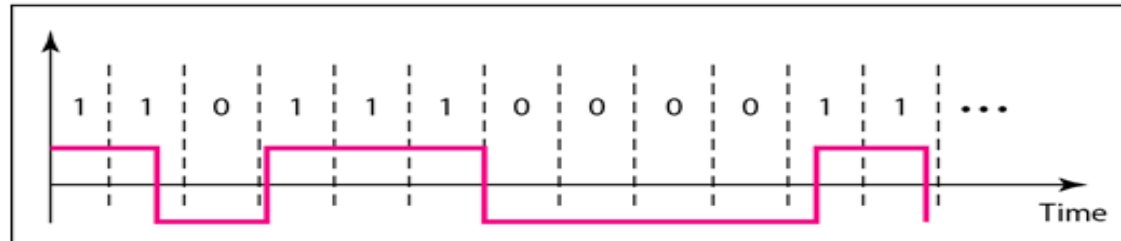
Considerations for choosing good coding scheme



- Self-synchronization
 - Digital signal includes timing information in the data being transmitted

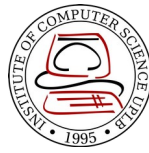


a. Sent



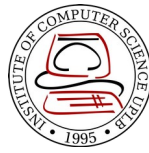
b. Received

Considerations for choosing good coding scheme



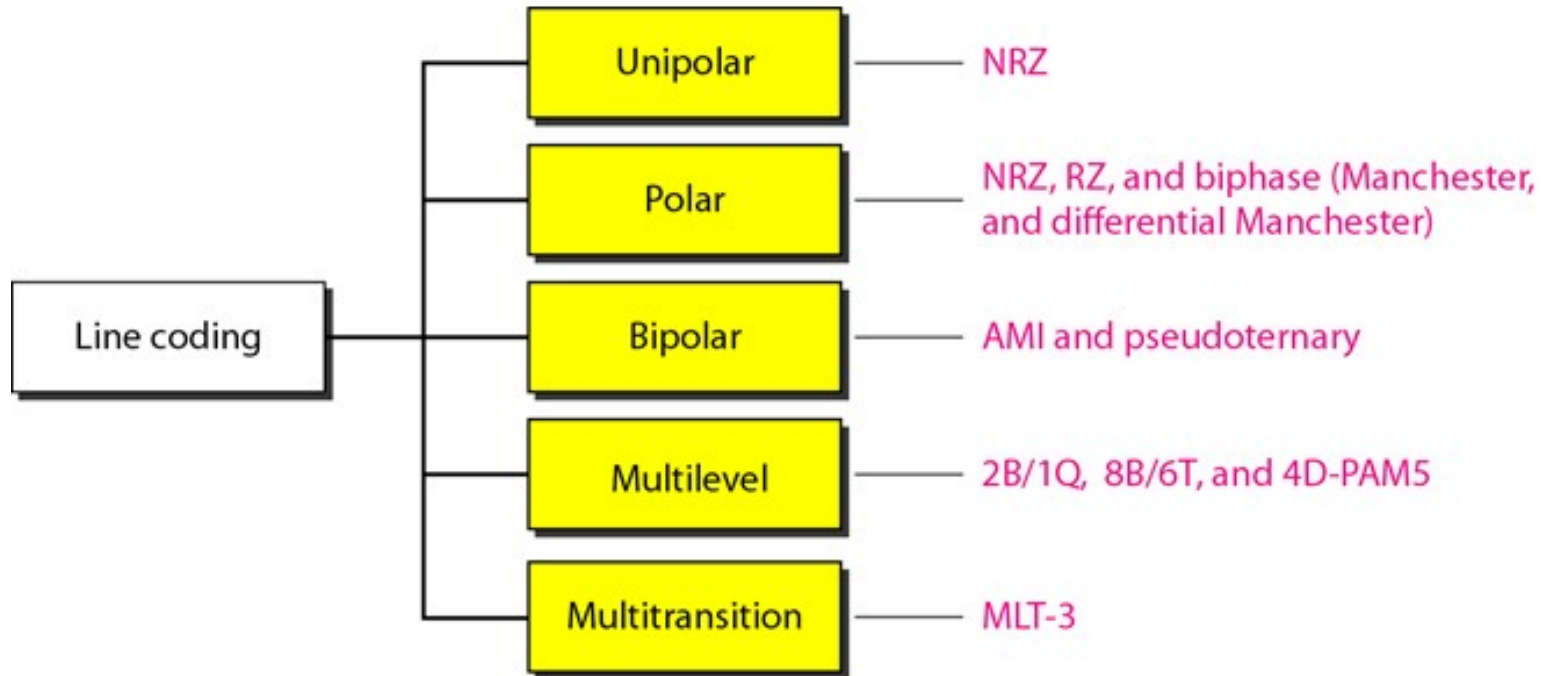
- Built-in Error Detection
 - Signal has built-in error detection
- Immunity to Noise and Interference
 - Signal generated is immune to noise
- Complexity
 - Scheme should be simple; the more robust and resilient the code, the more complex it is to implement and the price is often paid in baud rate or required bandwidth

Line Coding Schemes



- **Unipolar** – all signal levels are above or below time axis
- **Polar** – signal levels are on both sides of the time axis
- **Bipolar** – three levels: positive, negative, zero; one signal level alternates between positive and negative
- **Multilevel** – increase number of bits per baud
- **Multitransition** – more than 2 transition rules

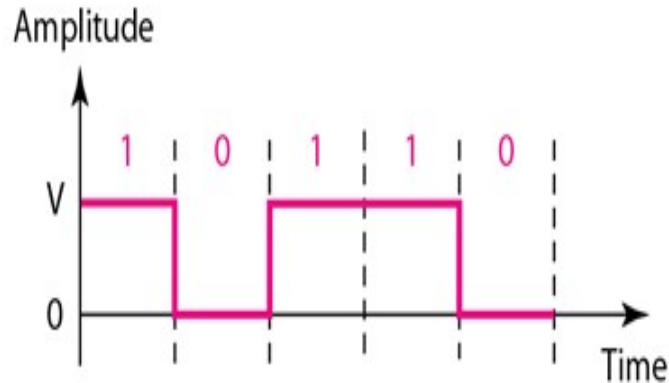
Line Coding Schemes



Unipolar NRZ



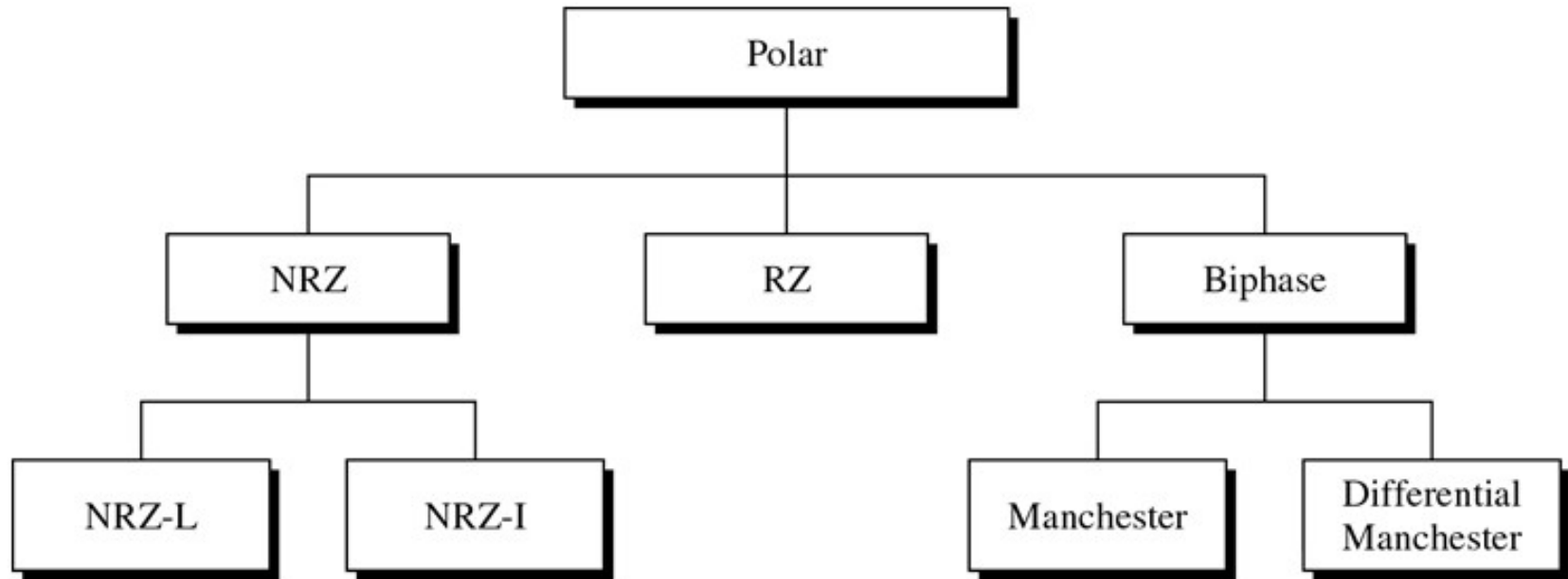
- Non-Return-to-Zero
- 1 – positive voltage, 0 – zero voltage
- Scheme is prone to baseline wandering and DC components. It has no synchronization or any error detection. It is simple but costly in power consumption



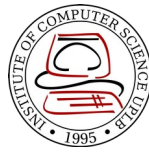
$$\frac{1}{2}V^2 + \frac{1}{2}(0)^2 = \frac{1}{2}V^2$$

Normalized power

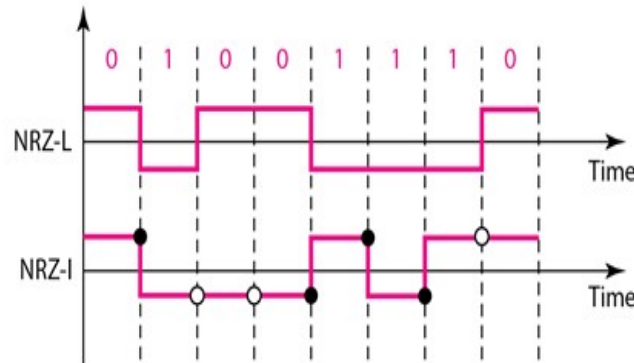
Polar



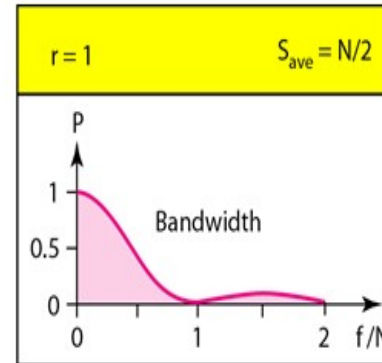
Polar NRZ



- NRZ-L (Level)
 - Level of voltage determines value of bit
- NRZ-I (Invert)
 - Change or lack of change in the level of the voltage determines the value of the bit (0 – no change)



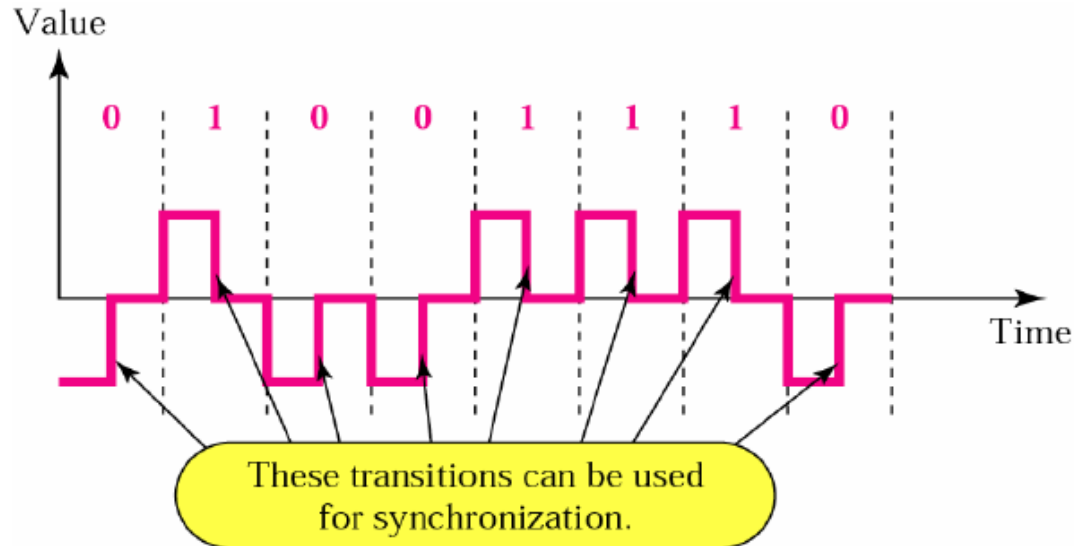
○ No inversion: Next bit is 0 ● Inversion: Next bit is 1



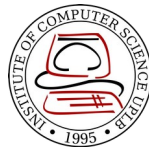
Polar RZ



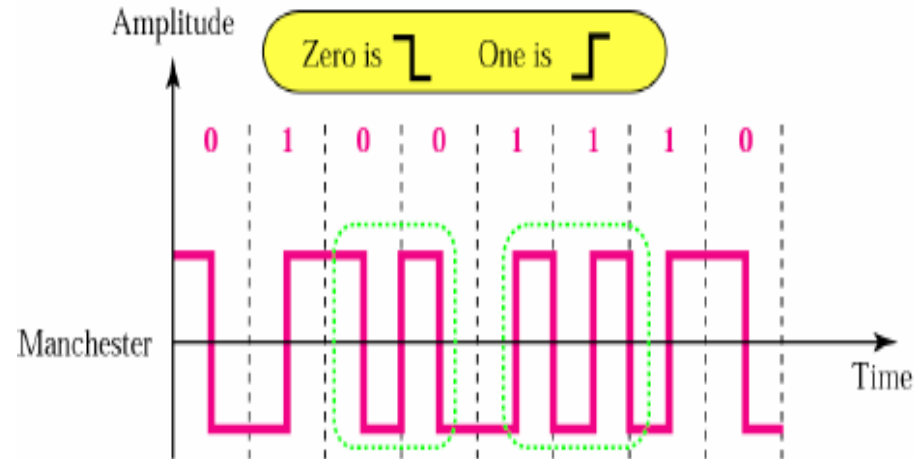
- Midbit transition, solves sync problem of NRZ
- More complex, more bandwidth, no DC component or baseline wandering



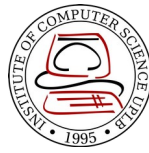
Polar Biphase



- Manchester
 - inversion at the middle of each bit interval is used for both synchronization and bit representation
 - 0 = positive to negative transition, 1 = negative to positive transition
 - fine for alternating sequence of bits (10101) but wastes bandwidth for long runs of 1s or 0s
 - Used by IEEE 802.3 (Ethernet)

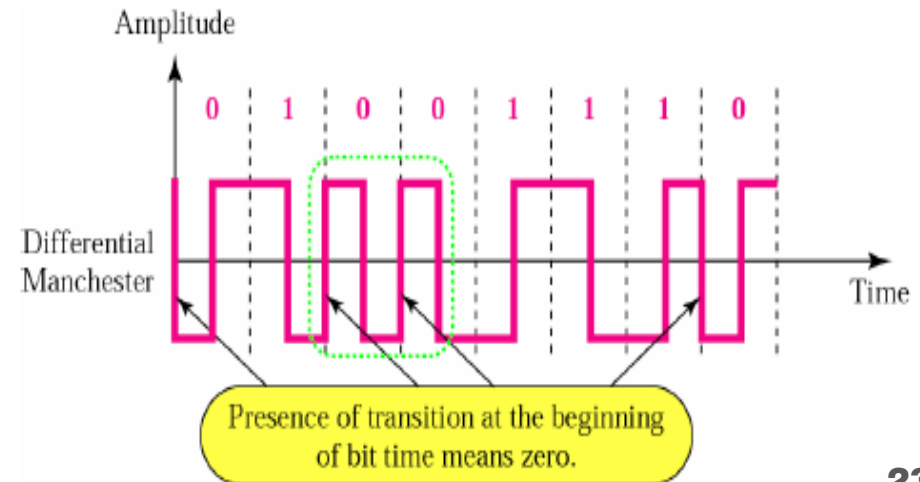


Polar Biphase



- Differential Manchester

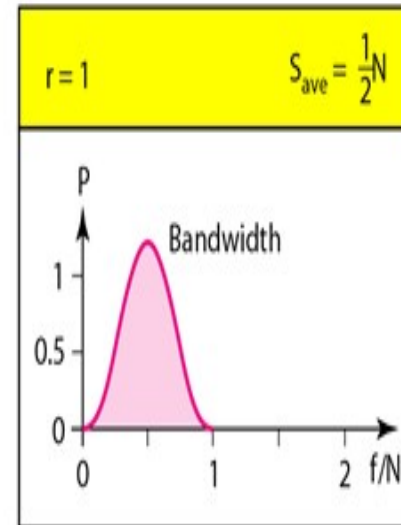
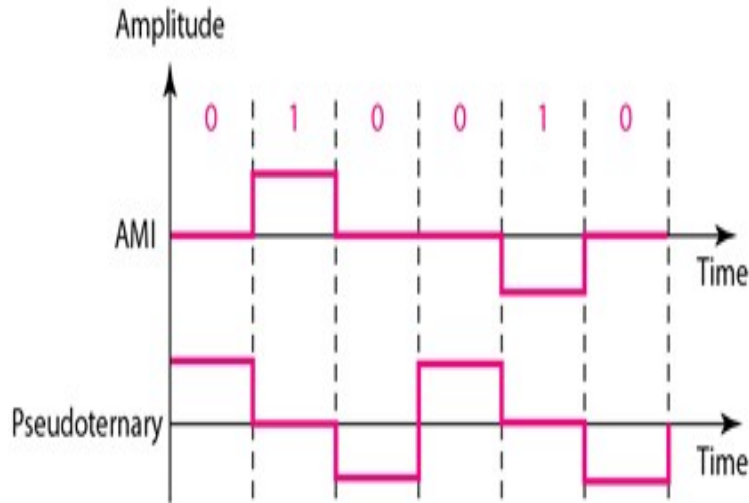
- inversion at the middle of each bit interval is used for synchronization, presence/absence of additional transition at the beginning of next bit interval identifies the bit
- fine for long runs of 1s but wastes bandwidth for long runs of 0s
- 0 = transition, 1 = no transition
- Used by IEEE 802.5 (Token Ring)



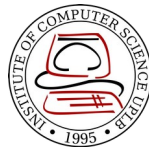
Bipolar



- **Alternate Mark Inversion (AMI)**
 - Neutral zero voltage – 0, alternating (+) and (-) - 1
- **Pseudoternary** – reverse of AMI



Multilevel Schemes



- Encode a pattern of m data elements into a pattern of n signal elements
- If $2^m > L^n$, we cannot represent data elements since not enough signals
- $mBnL$, pattern of m data elements is encoded in a pattern of n signal elements in which $2^m \leq L^n$
- $L = B$ for binary, T for ternary (3 levels), Q for quaternary (4 levels)

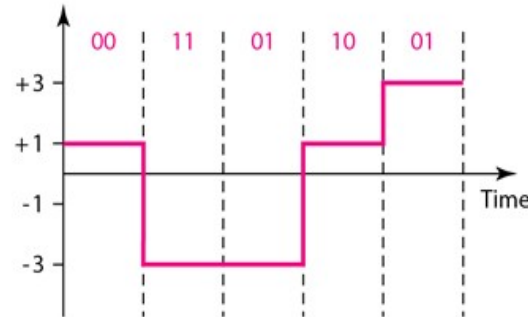
2B1Q



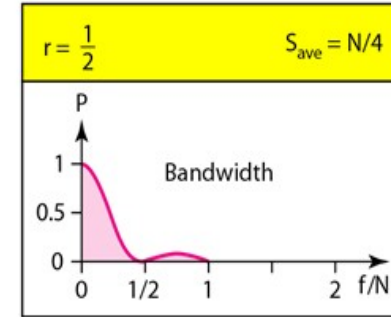
- Two binary, one quaternary
- Used in DSL (Digital Subscriber Line)

Next bits	Previous level: positive	Previous level: negative
	Next level	Next level
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

Transition table



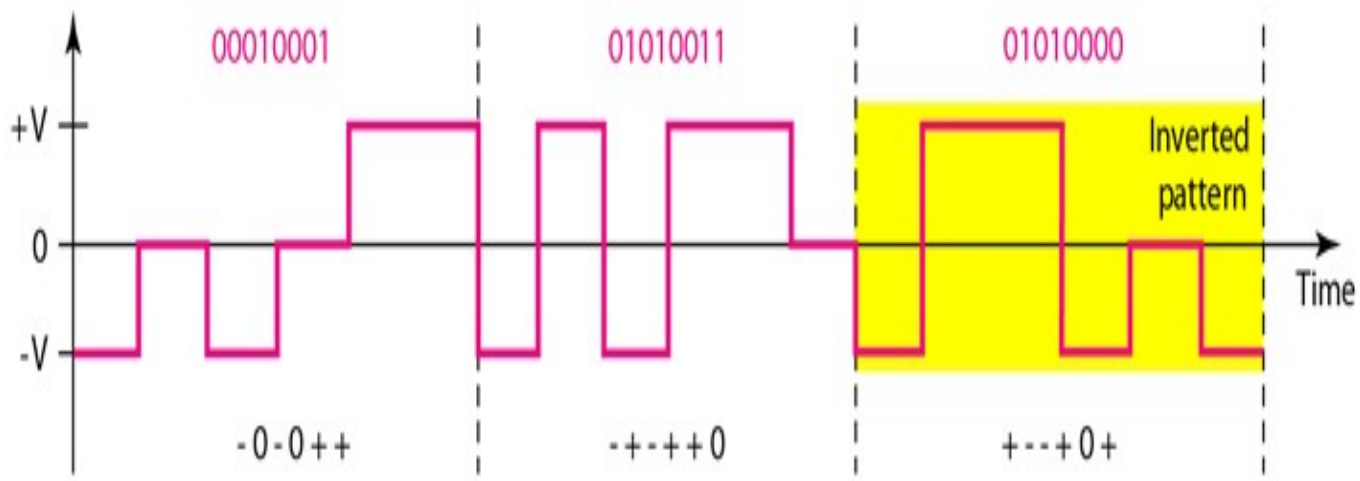
Assuming positive original level



8B6T



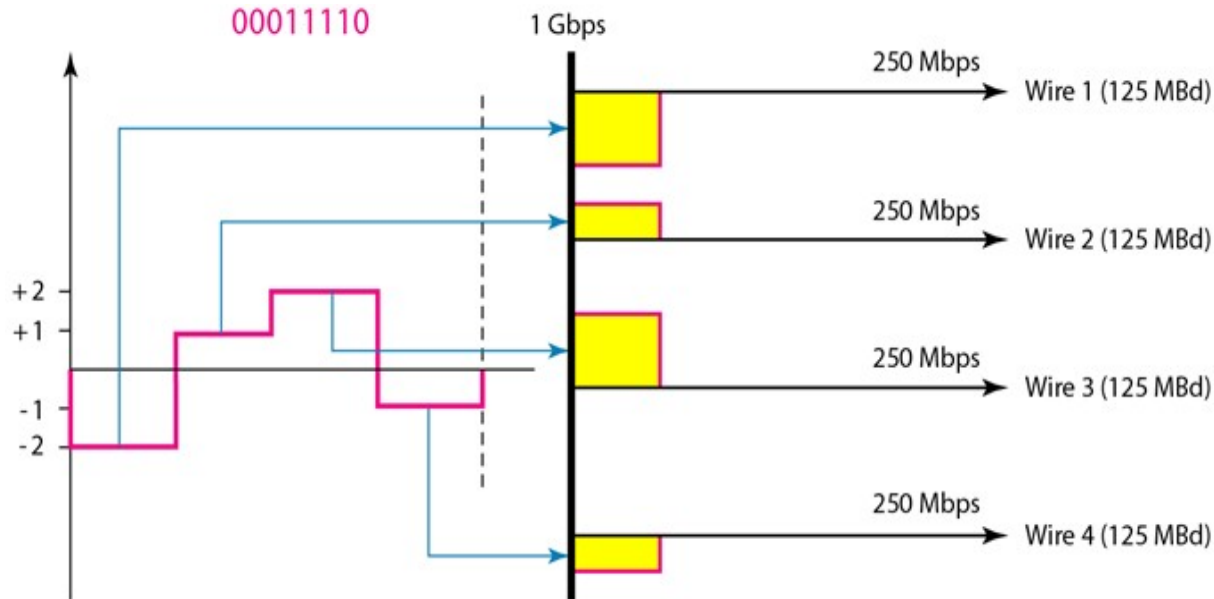
- Eight binary, six ternary
- Used in 100Base-4T, bandwidth $6N/8$



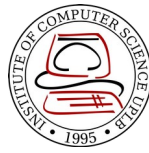
4D-PAM5



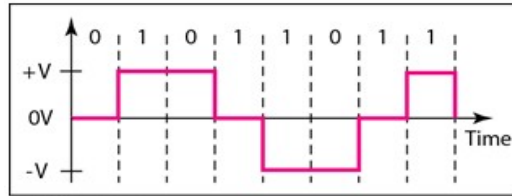
- Four dimensional five-level pulse amplitude modulation
- Used in Gigabit LANs, uses 4 wires



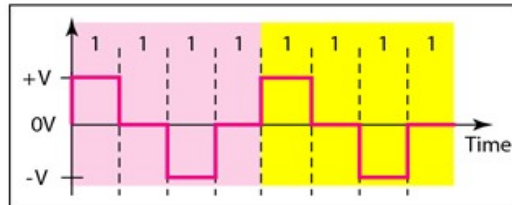
Multitransition: MLT3



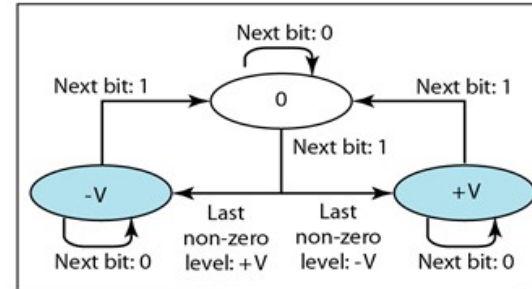
- Uses 3 levels ($+V$, 0 , $-V$)
- Uses 3 transition rules to move between levels
 - If the next bit is 0 , there is no transition
 - If the next bit is 1 and the current level is not 0 , the next level is 0
 - If the next bit is 1 and the current level is 0 , the next level is the opposite of the last nonzero level



a. Typical case

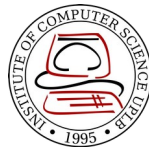


b. Worse case

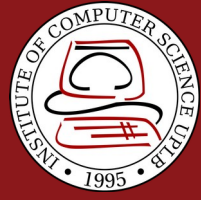


c. Transition states

Line Coding Summary



<i>Category</i>	<i>Scheme</i>	<i>Bandwidth (average)</i>	<i>Characteristics</i>
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Polar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multitransition	MLT-3	$B = N/3$	No self-synchronization for long 0s

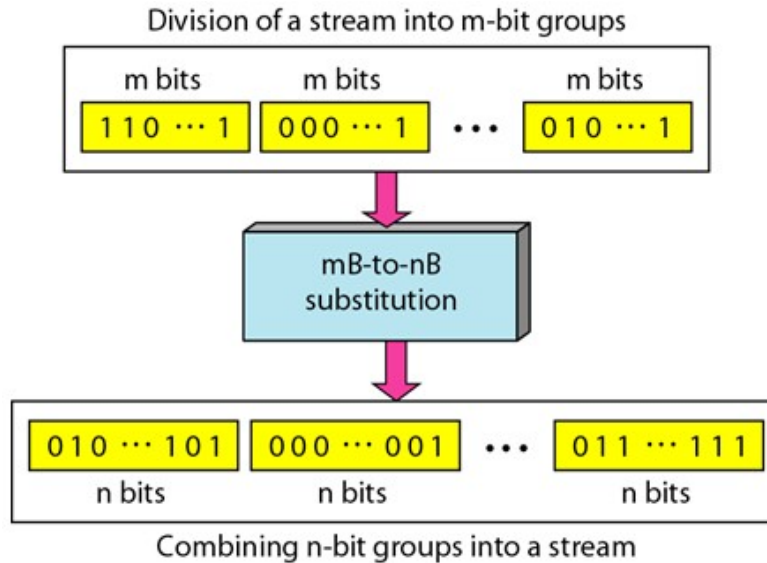


Block Coding

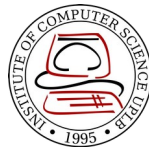
Block Coding



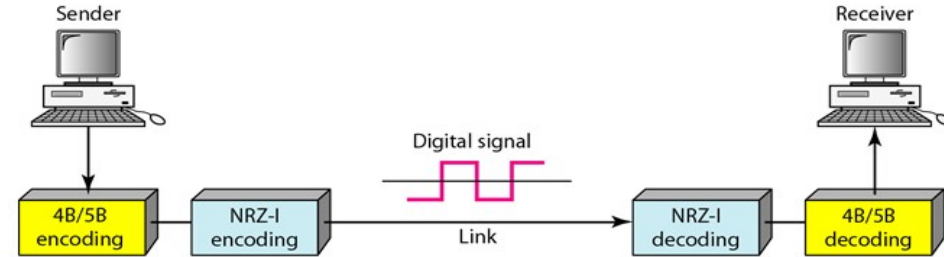
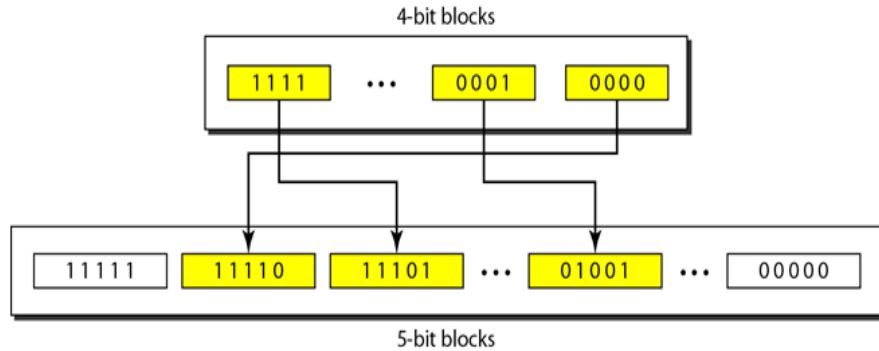
- Referred to as mB/nB coding; it replaces each m -bit group with n -bit group
- Redundant bit(s) are added to ensure synchronization and error detection
- Involves 3 steps: *division*, *substitution* and *combination*



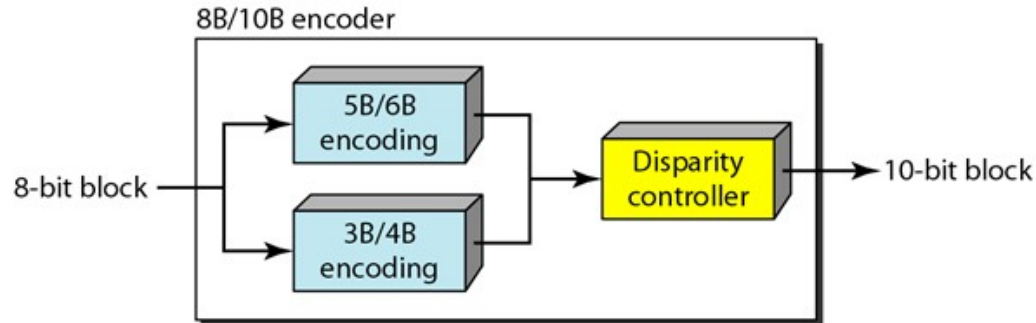
Block Coding Sample

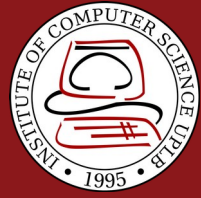


4B/5B (often combined with NRZ-I), used in FDDI



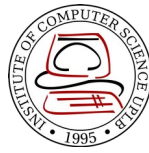
8B/10B





Scrambling

Scrambling



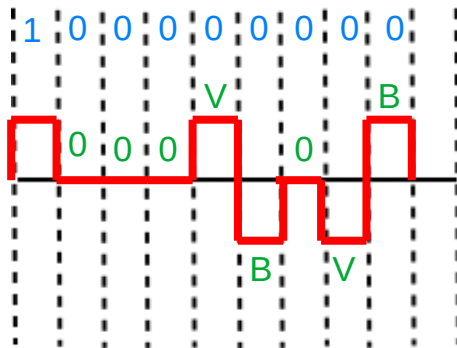
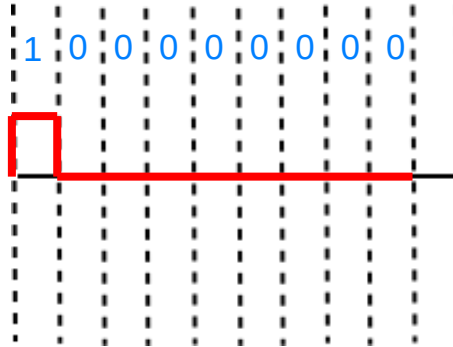
- The best code is one that does not increase the bandwidth for synchronization and has no DC components
- Bipolar AMI is most suitable for long distance transmission but long sequence of 0s disrupts synchronization
- Solution: *substitute long sequences of 0s with a combination of other levels*
- It is implemented at the same time as encoding

- Bipolar with 8-zero substitution
- Eight consecutive zero-level voltages are replaced by a sequence **000VB0VB**
 - **V** – non-zero voltage violating the AMI rule
 - **B** – bipolar non-zero voltage
- Does not change the bit rate, balances (+) and (-) voltages, substitution may change the polarity of a 1

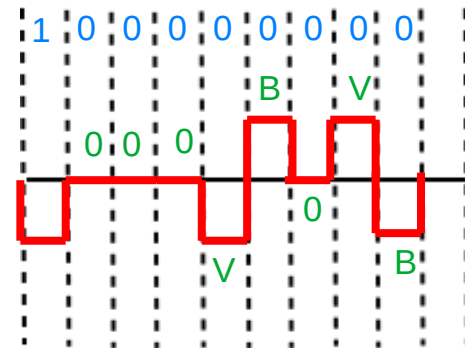
B8ZS



Without scrambling



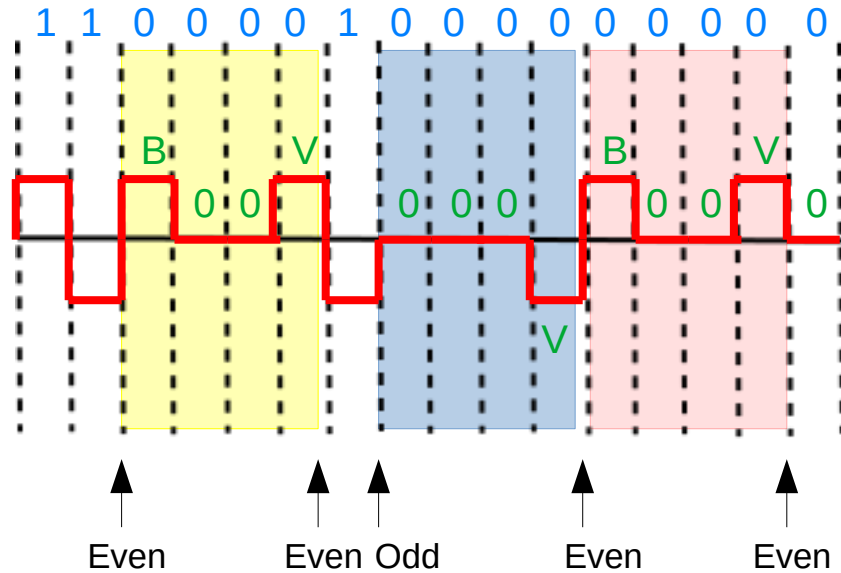
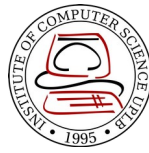
Previous level is positive

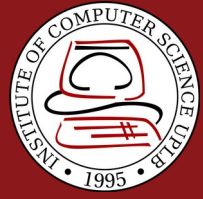


Previous level is negative

- High-density bipolar 3-zero
- Four consecutive zero-level voltages are replaced with a sequence of 000V or B00V
- Why two sequences? *To maintain the even number of non-zero pulses after each substitution*
 - If the number of nonzero pulses after the last substitution is odd, 000V
 - If the number of nonzero pulses after the last substitution is even, B00V

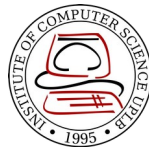
HDB3



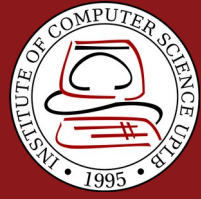


Analog to Digital Conversion

Analog to Digital Conversion



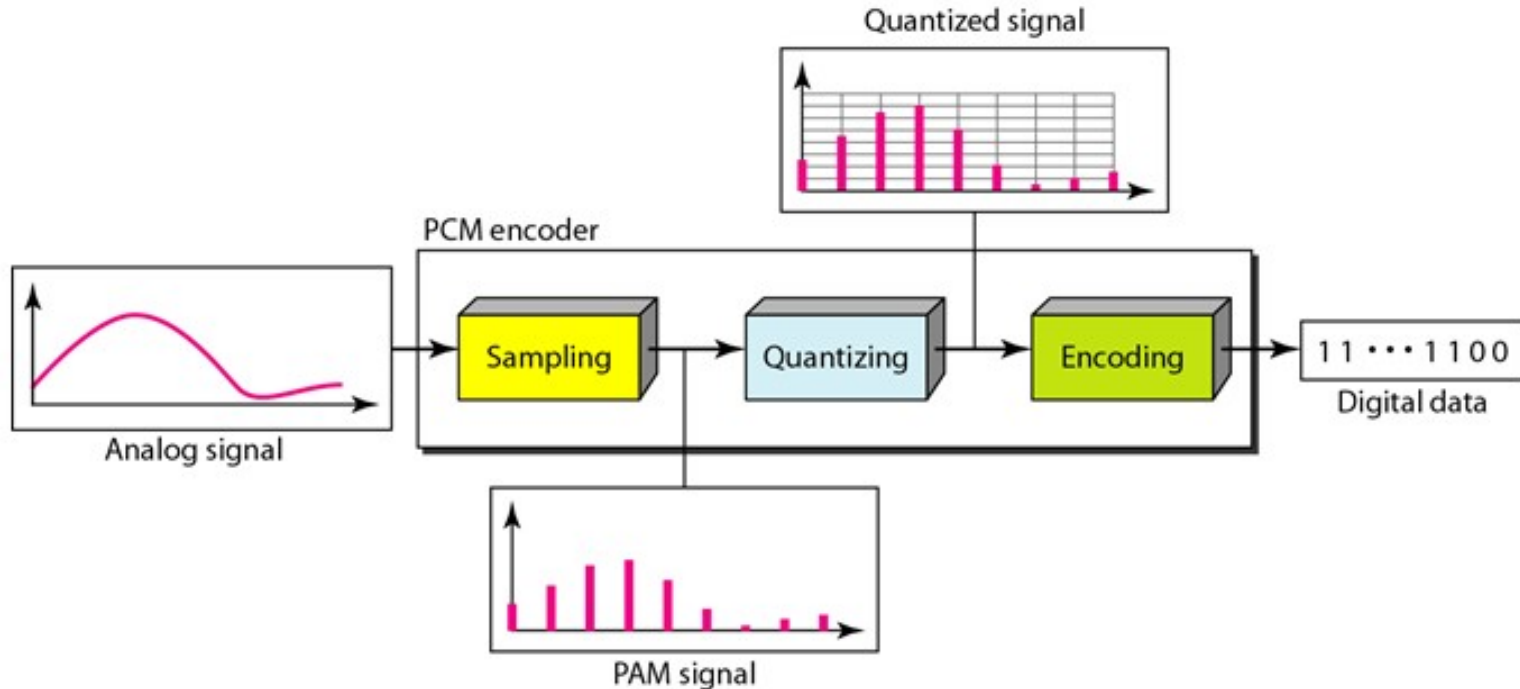
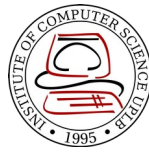
- Digital signal is superior to an analog signal because it is more robust to noise and can easily be recovered, corrected and amplified
- Tendency today is to change analog signal to digital data ([digitization](#))
- 2 common techniques:
 - [Pulse Code Modulation](#) (PCM)
 - [Delta Modulation](#)



Pulse Code Modulation

- Consists of 3 steps to digitize an analog signal
 - 1) Sampling
 - 2) Quantization
 - 3) Binary encoding
- Before we sample, we have to filter the signal to limit the maximum frequency of the signal as it affects the sampling rate
- Filtering should ensure that we do not distort the signal, ie remove high frequency components that affect the signal shape

PCM Encoder Components



Sampling

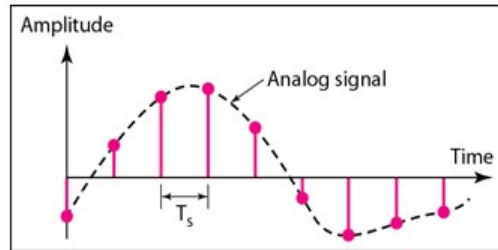


- Analog signal is sampled within a certain time period (sampling period or **sampling interval** T_s)
- Inverse of sampling period is called sampling rate or **sampling frequency** f_s
- Process is referred to as **Pulse Amplitude Modulation** (PAM) and the outcome is a signal with analog (non-integer) values
- Sampling rate must be **at least twice the highest frequency** in the original signal (Nyquist theorem)

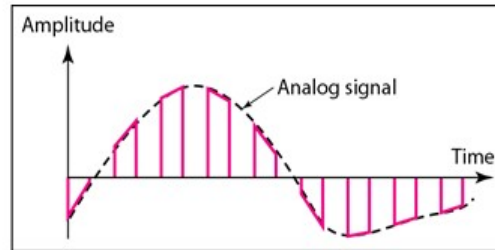
Sampling Methods



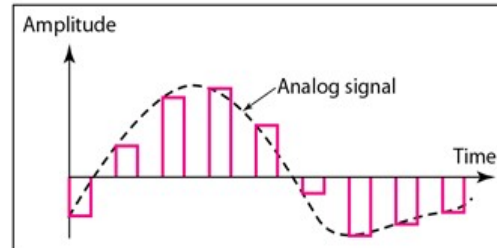
- **Ideal** – an impulse at each sampling instant
- **Natural** – a pulse short width with varying amplitude
- **Flat-top** – sample and hold, like natural with single amplitude value



a. Ideal sampling

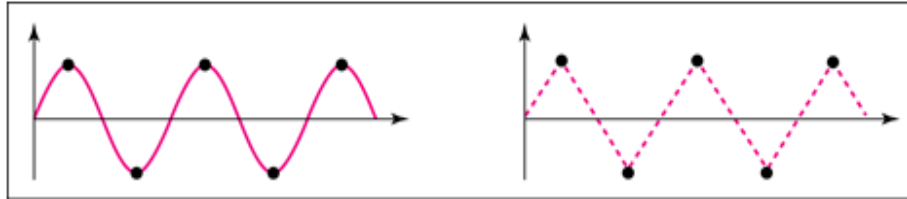


b. Natural sampling



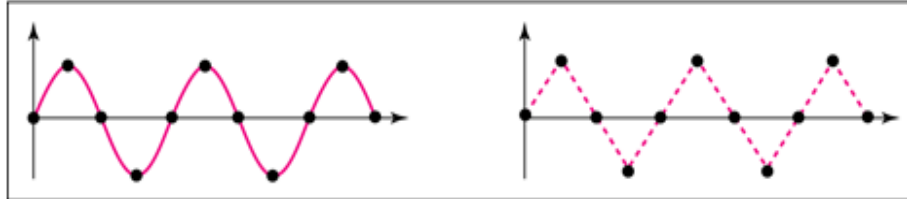
c. Flat-top sampling

Sampling Rate



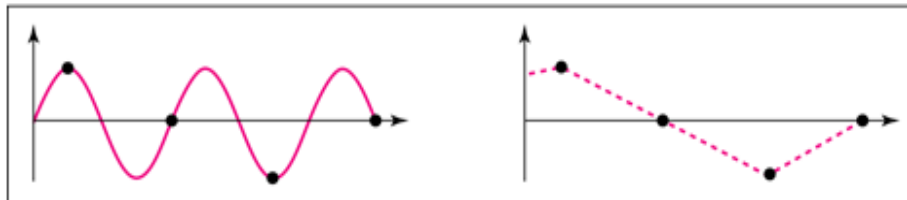
a. Nyquist rate sampling: $f_s = 2f$

Sampling at the Nyquist rate can create a good approximation of the original sine wave.



b. Oversampling: $f_s = 4f$

Oversampling can also create the same approximation, but is redundant and unnecessary.



c. Undersampling: $f_s = f$

Sampling below the Nyquist rate does not produce a signal that looks like the original sine wave.

Quantization



- Sampling results in a series of pulses of varying amplitude values ranging between two limits: $v_{\min}$ and $v_{\max}$
- The amplitude values are infinite between the two limits and it needs to be mapped onto a finite set of known values
- Divide the range into L zones, each of height delta

$$\text{delta} = (v_{\max} - v_{\min}) / L$$

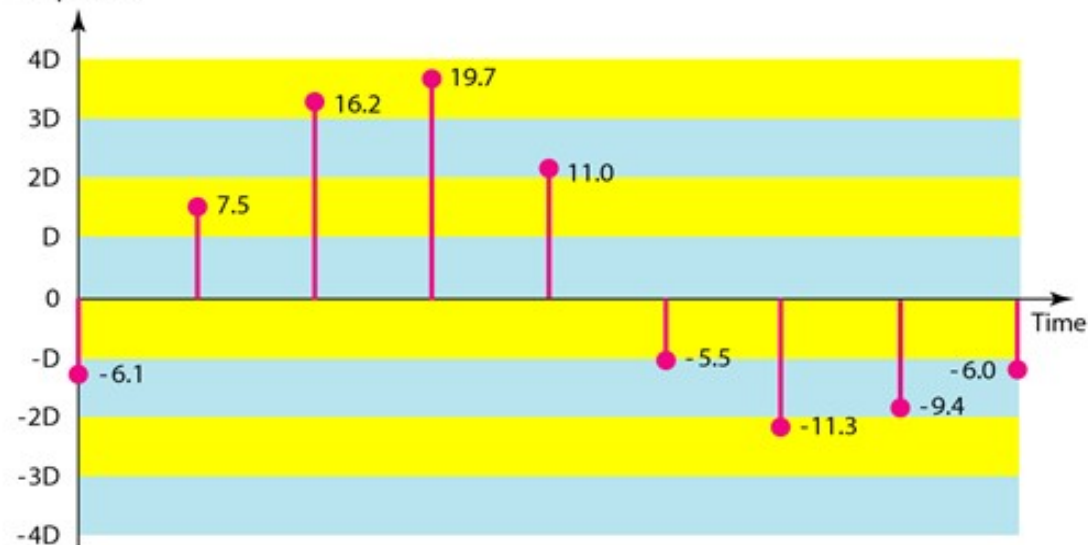
- Assign quantized values of 0 to $L-1$ to the midpoint of each zone
- Approximate the value of the sample amplitude to the quantized values

Quantization and Encoding

Quantization codes

7
6
5
4
3
2
1
0

Normalized amplitude



Normalized PAM values

-1.22 1.50 3.24 3.94 2.20 -1.10 -2.26 -1.88 -1.20

Normalized quantized values

-1.50 1.50 3.50 3.50 2.50 -1.50 -2.50 -1.50 -1.50

Normalized error

-0.38 0 +0.26 -0.44 +0.30 -0.40 -0.24 +0.38 -0.30

Quantization code

2 5 7 7 6 2 1 2 2

Encoded words

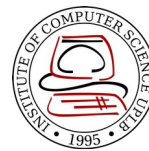
010 101 111 111 110 010 001 010 010

Assume $v_{\max} = +20V$ and $v_{\min} = -20V$

$L = 8$

$$\Delta = \frac{20V - (-20V)}{8} = 5$$

Encoding



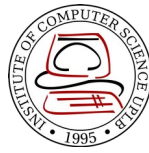
- Each zone is assigned a binary code
- Number of bits required to encode the zones is obtained as follows:

$$n_b = \log_2 L$$

- Given on the previous example

$$n_b = \log_2 8 = 3$$

Bit Rate and Bandwidth Requirement

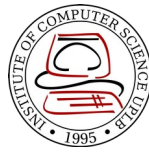


- Bit rate of a PCM signal can be calculated by multiplying the number of bits per sample and sampling rate:

$$\text{bit rate} = n_b \times f_b$$

- Bandwidth required to transmit this signal depends on the type of line encoding used. A digitized signal will always need more bandwidth than the original analog signal

Question



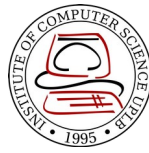
Question: What is the bit rate needed to digitize human voice assuming 8 bits per sample?

Human voice contains frequency between 0 to 4000Hz

$$f_s = 4000 \times 2 = 8000 \text{ samples/s}$$

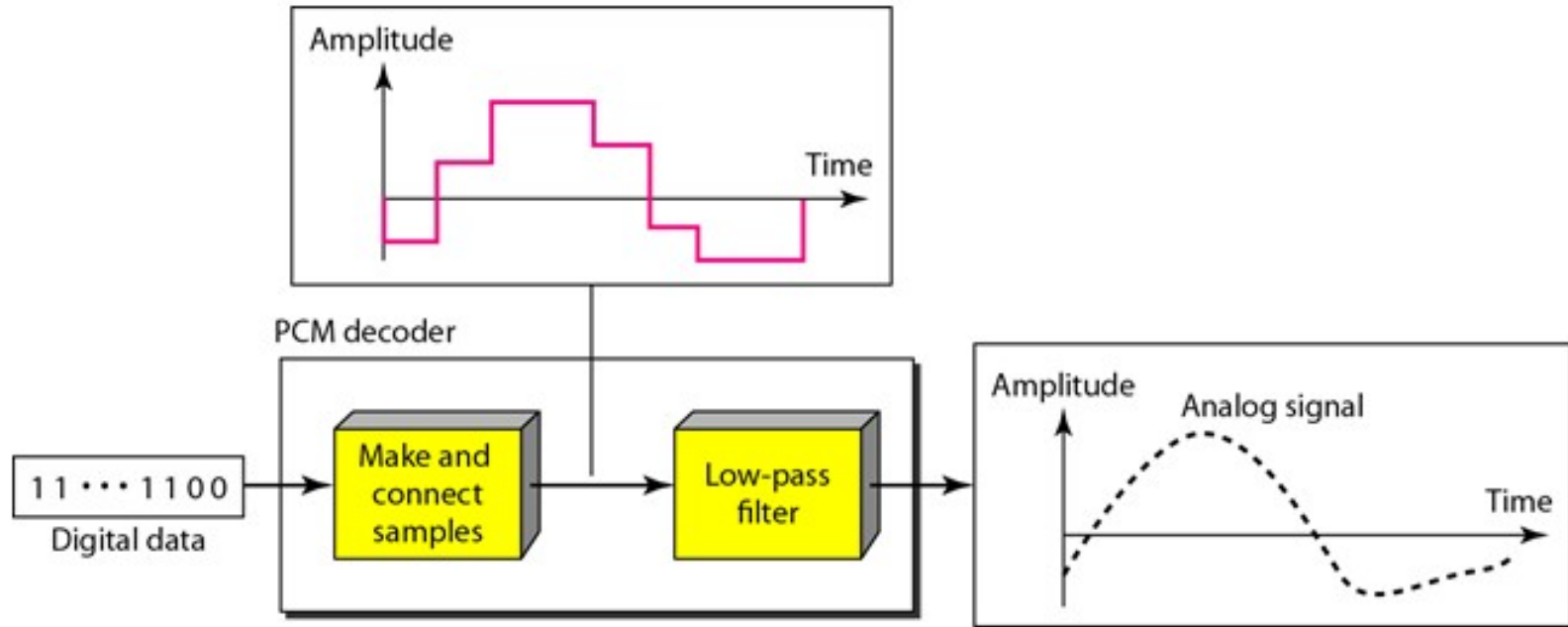
$$\text{bit rate} = 8 \times 8000 = 64,000 \text{ bps} = 64\text{kbps}$$

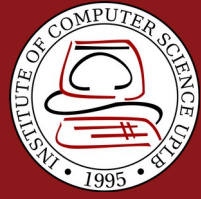
PCM Decoder



- To recover an analog signal from a digitized signal:
 - Use a circuit that holds the amplitude value of a pulse till the next pulse arrives
 - Pass this signal through a low pass filter with cutoff frequency that is equal to the highest frequency in the pre-sampled signal
- The higher the value of L , the less distorted a signal is recovered

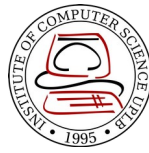
PCM Decoder



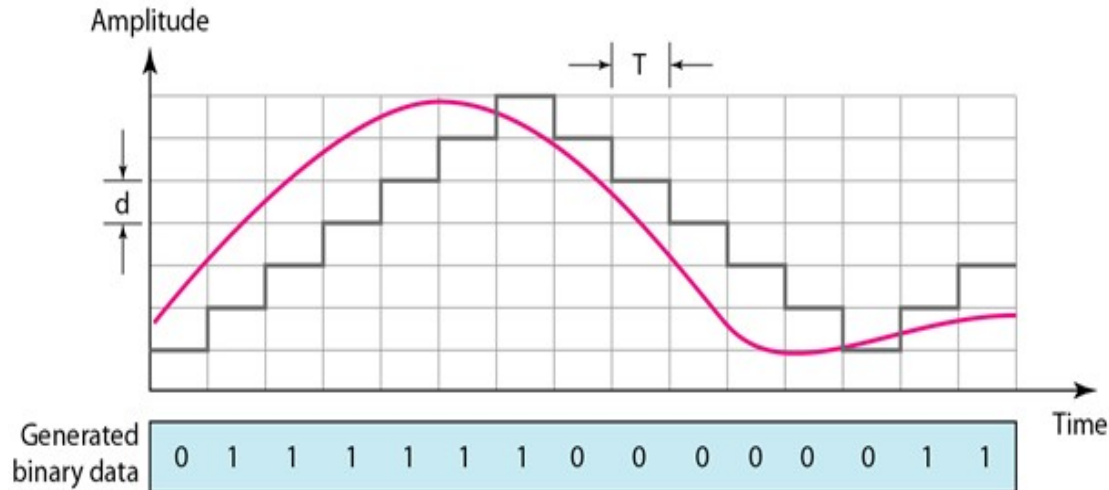


Delta Modulation

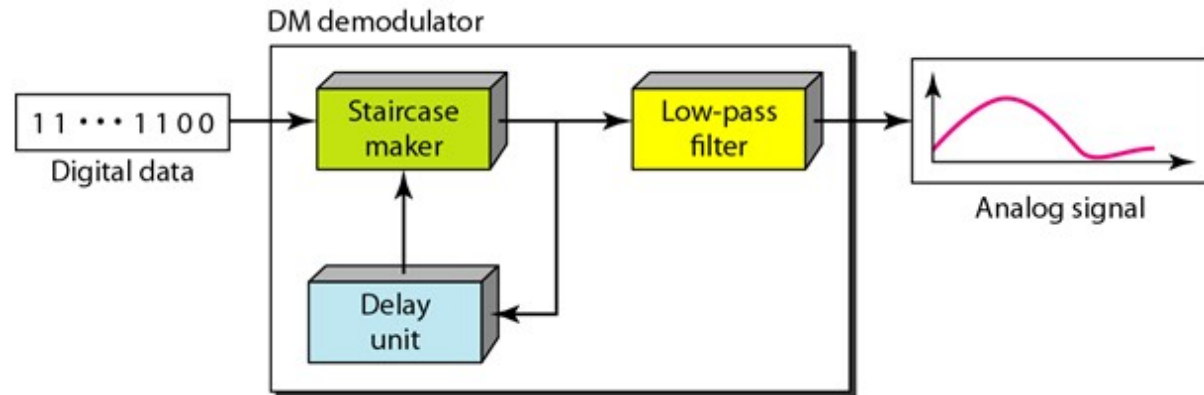
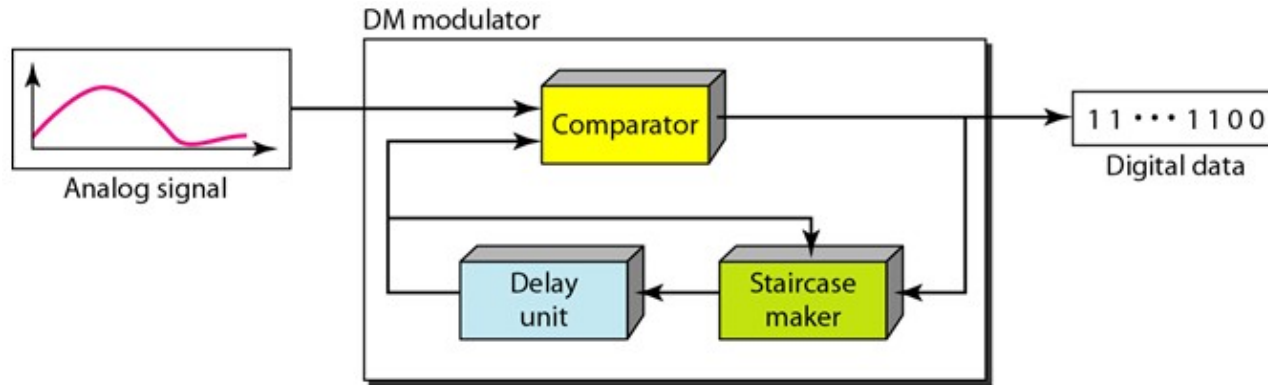
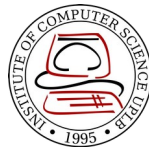
Delta Modulation

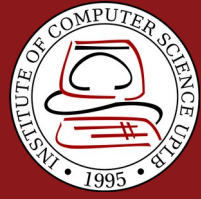


- This scheme finds the change from the previous sample
- If pulse at time t_{n+1} is higher in amplitude than the pulse at t_n , a single bit (assume "1") is used to indicate positive value; if pulse is lower, it results in a negative value ("0")



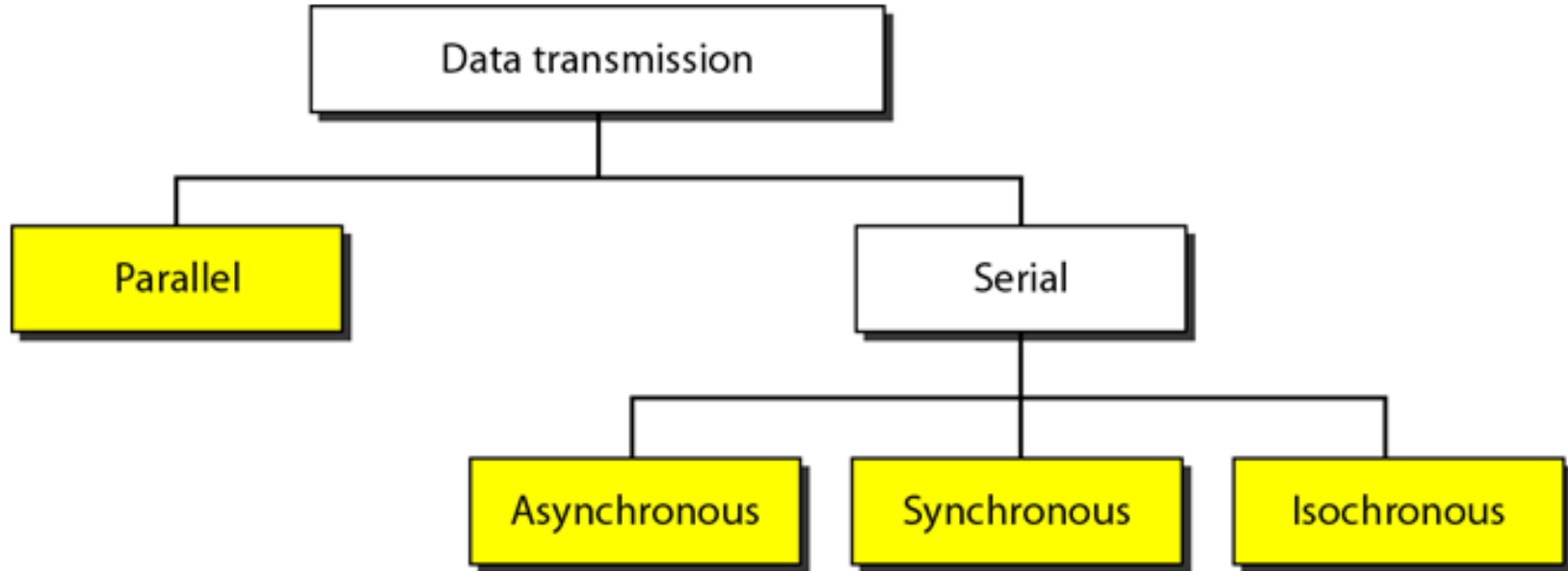
DM Components





Transmission Modes

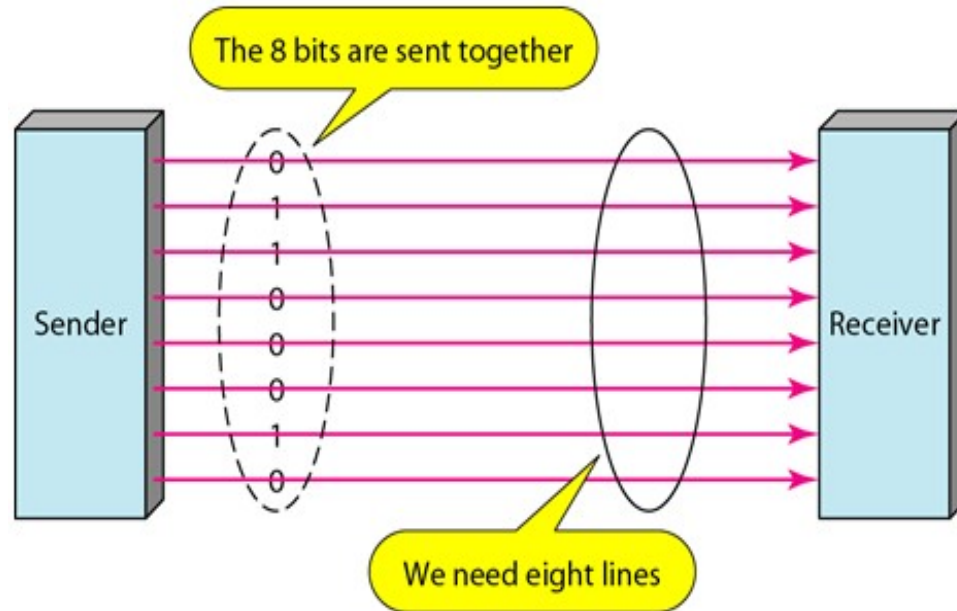
Transmission Modes



Parallel Transmission



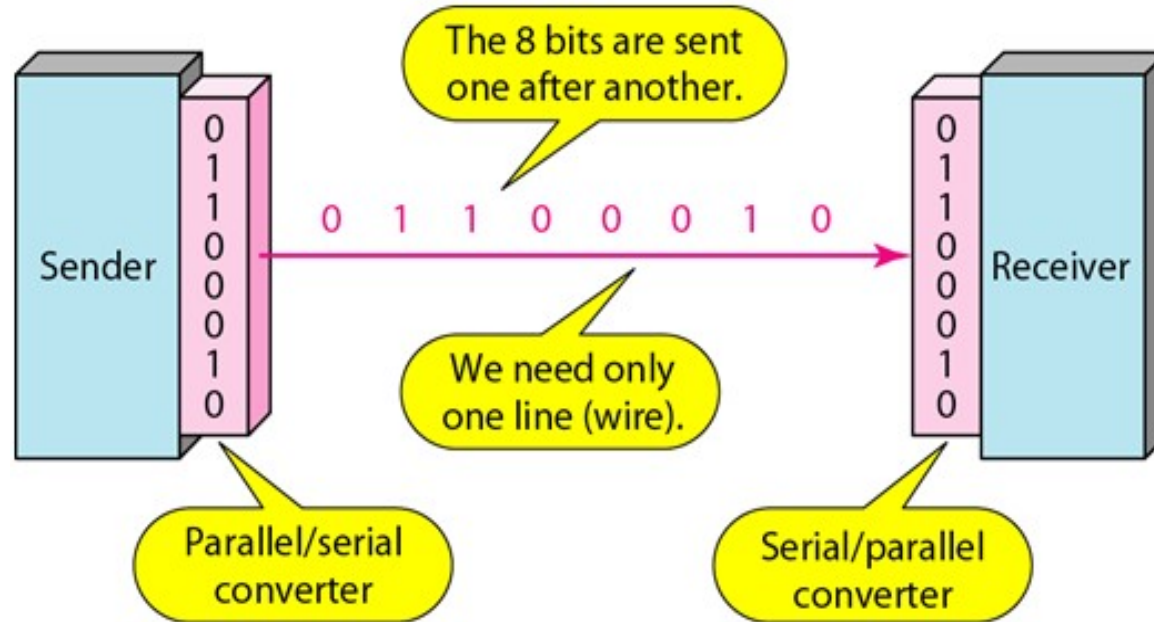
- Use n wires to send n bits of data



Serial Transmission



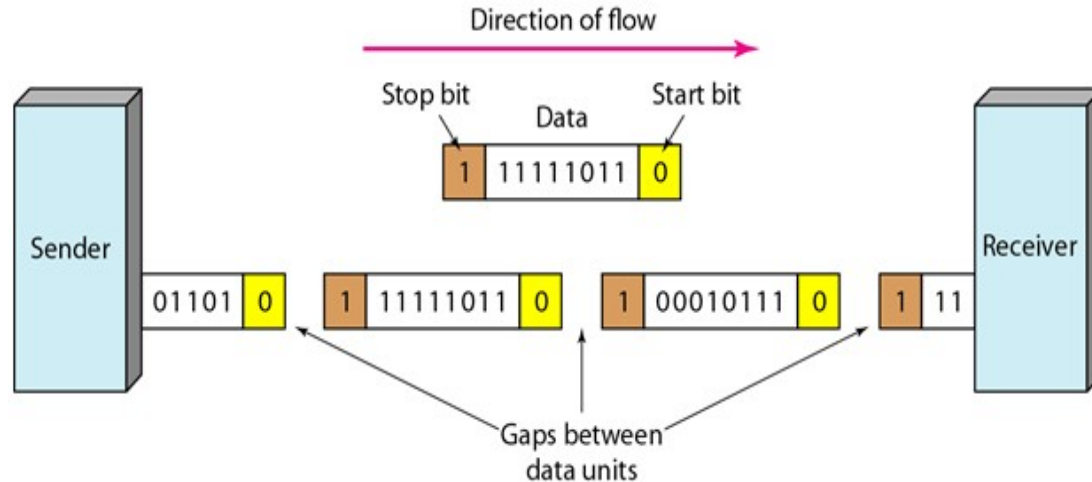
- 1 bit at a time



Asynchronous Transmission



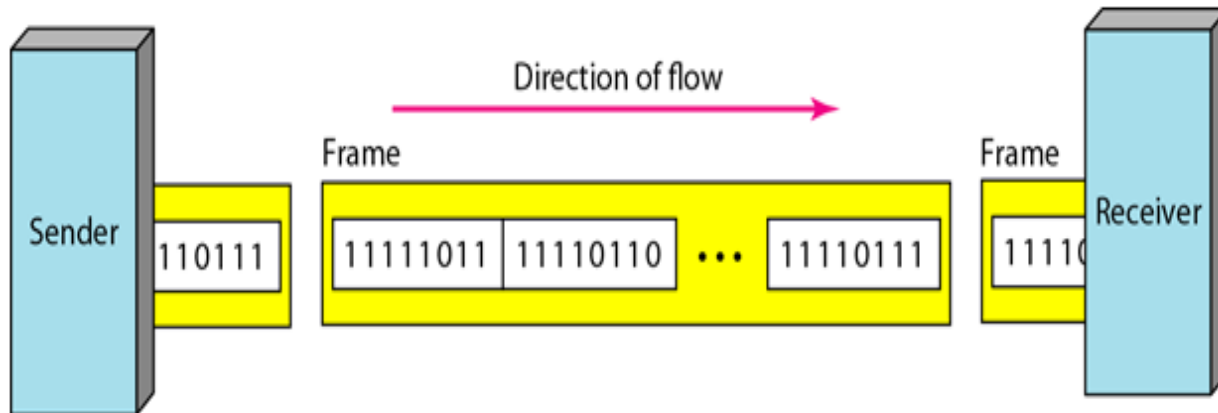
- Timing of a signal is unimportant
- Uses “*patterns*” - grouping of bit streams into bytes
- **Start bit** (0), **Stop bit** (1), may have gap in-between
- No synchronization at byte level, but present in bit level
- Bits can be transferred at random intervals and the data rate of the stream is not required to be constant



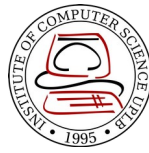
Synchronous Transmission



- Bit stream combined into frames which may contain multiple bytes
- The receiver is responsible for decoding
- Byte synchronization is accomplished in the data link layer
- Bits must be transferred in sync with a clock signal



Isochronous Transmission



- Combines the features of an asynchronous and synchronous data transfer system
- Transmission of bits is fixed with equal gaps
- Each transmission begins with a start packet
- Used in multimedia data

References



- Bucaro, S. Synchronous, Asynchronous, Isochronous. What Does It Mean? Retrieved from <http://bucarotechelp.com/networking/basics/88092302.asp>
- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.